ZOMATO DATA ANALYSIS PROJECT

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df=pd.read_csv('Zomato data 1.csv')
print(df.head())
```

```
                   name online_order book_table   rate  votes  \
0                 Jalsa          Yes        Yes  4.1/5    775
1        Spice Elephant          Yes         No  4.1/5    787
2       San Churro Cafe          Yes         No  3.8/5    918
3  Addhuri Udupi Bhojana           No         No  3.7/5     88
4         Grand Village           No         No  3.8/5    166

   approx_cost(for two people) listed_in(type)
0                          800          Buffet
1                          800          Buffet
2                          800          Buffet
3                          300          Buffet
4                          600          Buffet
```

In [3]:
```
df
```

Out[3]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1/5	775	800	Buffet
1	Spice Elephant	Yes	No	4.1/5	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8/5	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	Buffet
4	Grand Village	No	No	3.8/5	166	600	Buffet
...	...	...	...	...	...	...	...
143	Melting Melodies	No	No	3.3/5	0	100	Dining
144	New Indraprasta	No	No	3.3/5	0	150	Dining
145	Anna Kuteera	Yes	No	4.0/5	771	450	Dining
146	Darbar	No	No	3.0/5	98	800	Dining
147	Vijayalakshmi	Yes	No	3.9/5	47	200	Dining

148 rows × 7 columns

convert the datatype of column rate..

In [4]:

```
def handlerate(value):
    value=str(value).split('/')
    value=value[0]
    return float(value)

df['rate']=df['rate'].apply(handlerate)
print(df.head())
```

```
                     name online_order book_table  rate  votes  \
0                   Jalsa          Yes        Yes   4.1    775
1          Spice Elephant          Yes         No   4.1    787
2         San Churro Cafe          Yes         No   3.8    918
3  Addhuri Udupi Bhojana           No         No   3.7     88
4           Grand Village           No         No   3.8    166

   approx_cost(for two people) listed_in(type)
0                          800          Buffet
1                          800          Buffet
2                          800          Buffet
3                          300          Buffet
4                          600          Buffet
```

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   name                         148 non-null    object
 1   online_order                 148 non-null    object
 2   book_table                   148 non-null    object
 3   rate                         148 non-null    float64
 4   votes                        148 non-null    int64
 5   approx_cost(for two people)  148 non-null    int64
 6   listed_in(type)              148 non-null    object
dtypes: float64(1), int64(2), object(4)
memory usage: 8.2+ KB
```

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
name                           0
online_order                   0
book_table                     0
rate                           0
votes                          0
approx_cost(for two people)    0
listed_in(type)                0
dtype: int64
```

In [21]:
```
# ques....
```

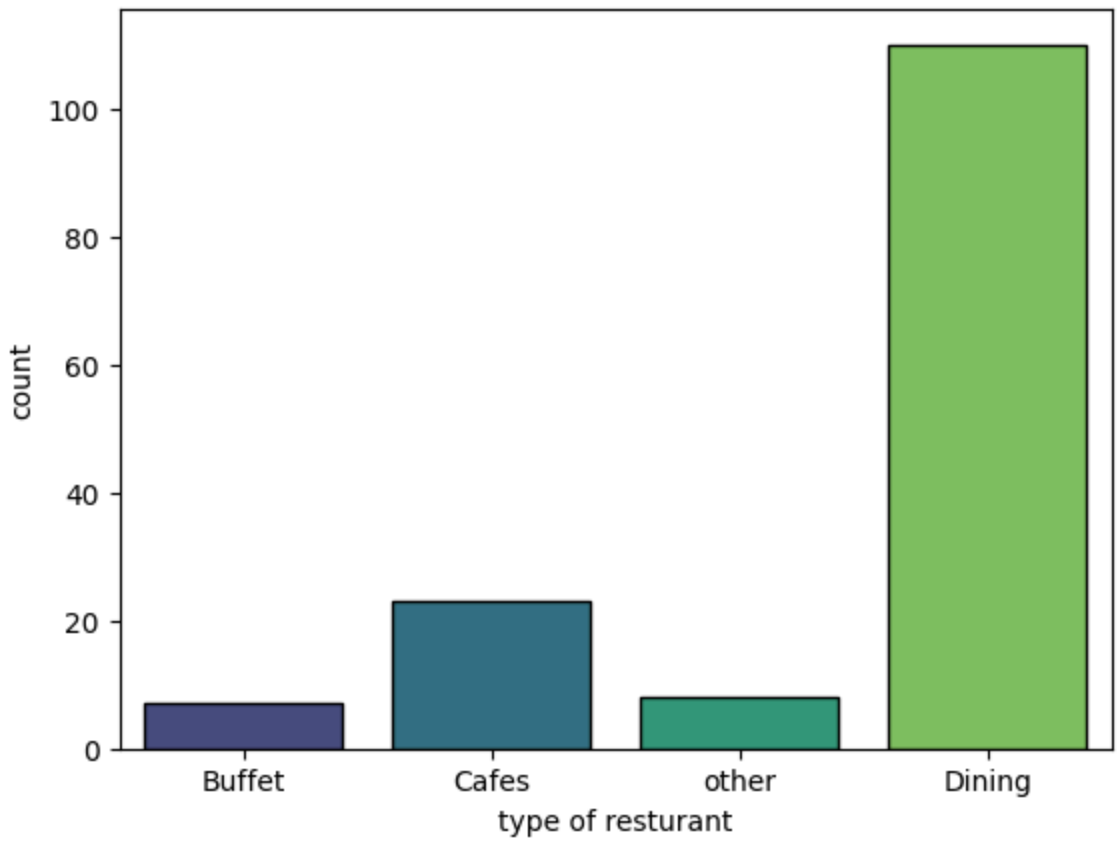
In [16]:
```
#  what type of restaurant do the majority of customers order from..
```

In [17]:
```
df.head(3)
```

Out[17]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet

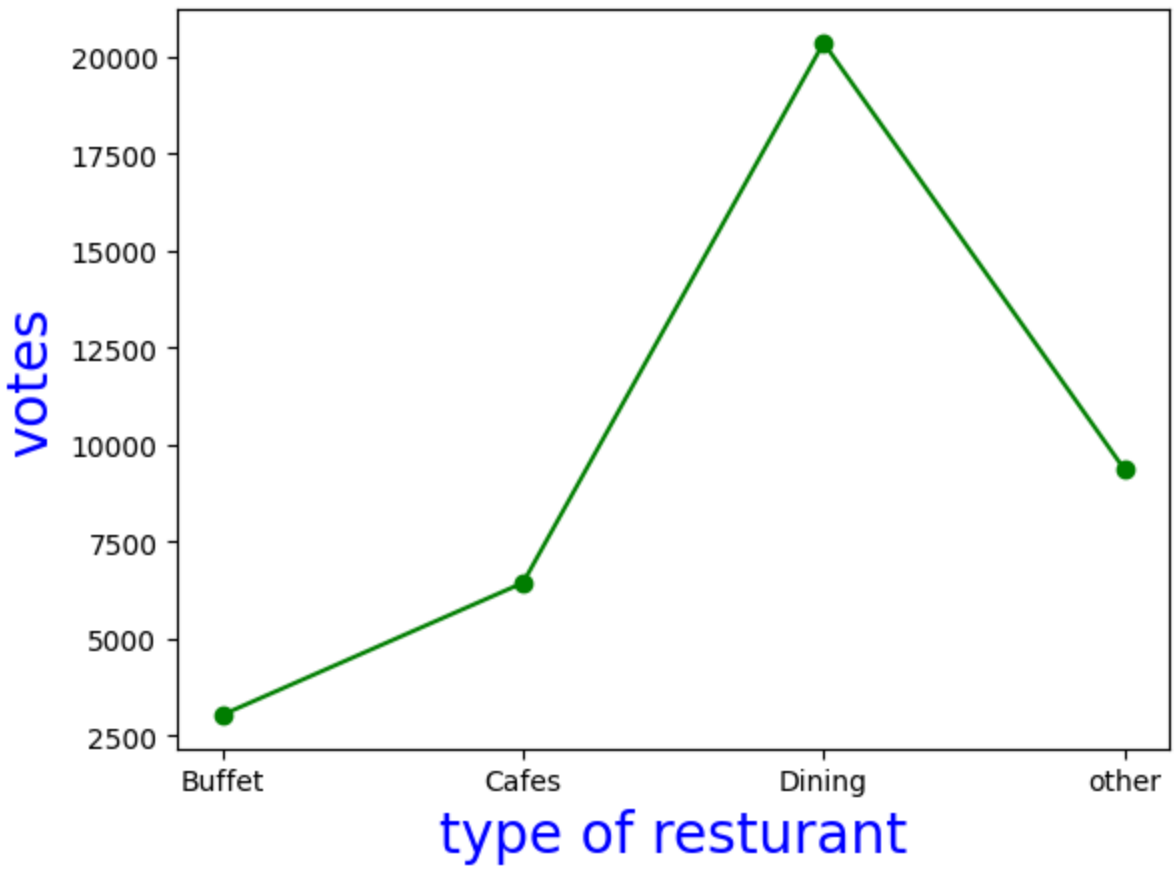
In [65]:
```
sns.countplot(x=df['listed_in(type)'],data=df,hue='listed_in(type)',palette='viridi
plt.xlabel("type of resturant")
plt.show()
```



In [20]:
```
#  conclusion: majority of customers order from the dining resturant.
```

In [22]:
```
# ques 2.. how many votes has each type of resturant received from customers..
```

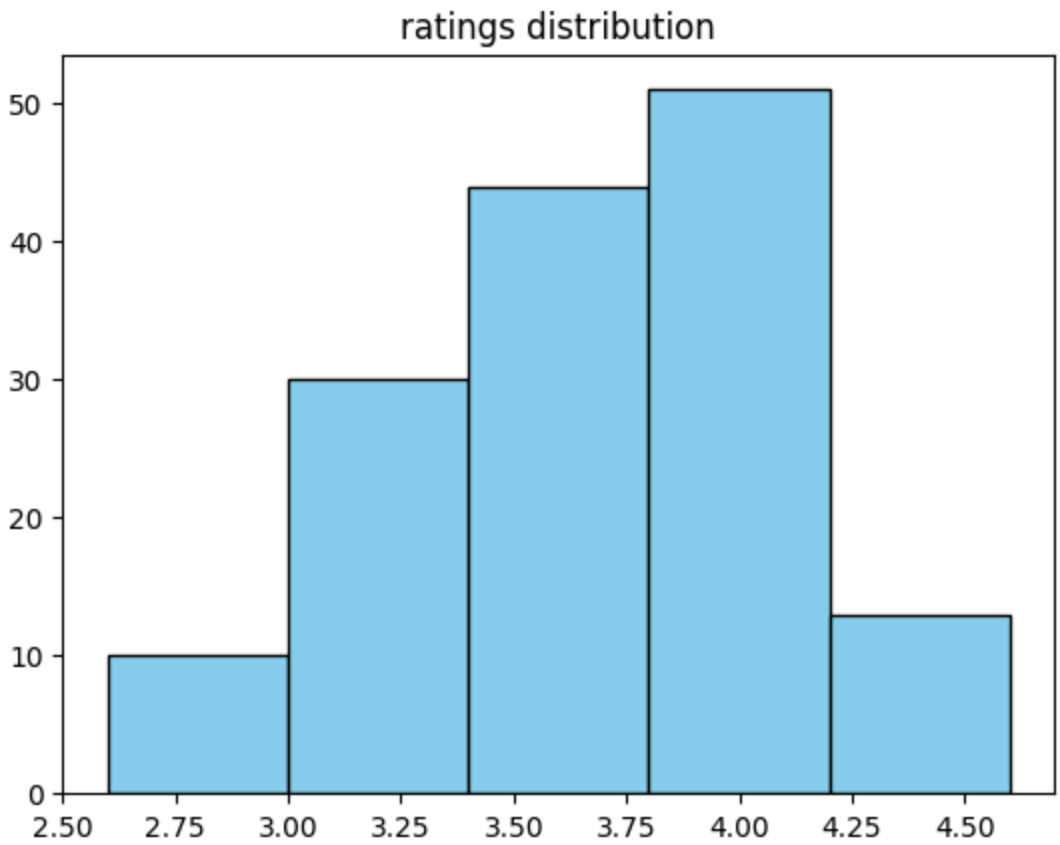
In [33]:
```
grouped_data=df.groupby('listed_in(type)')['votes'].sum()
result=pd.DataFrame({'votes': grouped_data})
plt.plot(result,c='green',marker='o')
plt.xlabel('type of resturant',c='blue',size=20)
plt.ylabel('votes',c='blue',size=20)
plt.show()
```



In [34]:
```
# conclusion.. dining resturant has received maximum votes.
```

In [7]:
```
# ques...3... what are the ratings that the majority of resturants have received...
```

In [67]:
```
plt.hist(df['rate'],bins =5,color='skyblue',edgecolor='black')
plt.title('ratings distribution')
plt.show()
```



In [12]:
```
# conclusion: the majority resturant receieved ratings from 3.5 to 4.
```

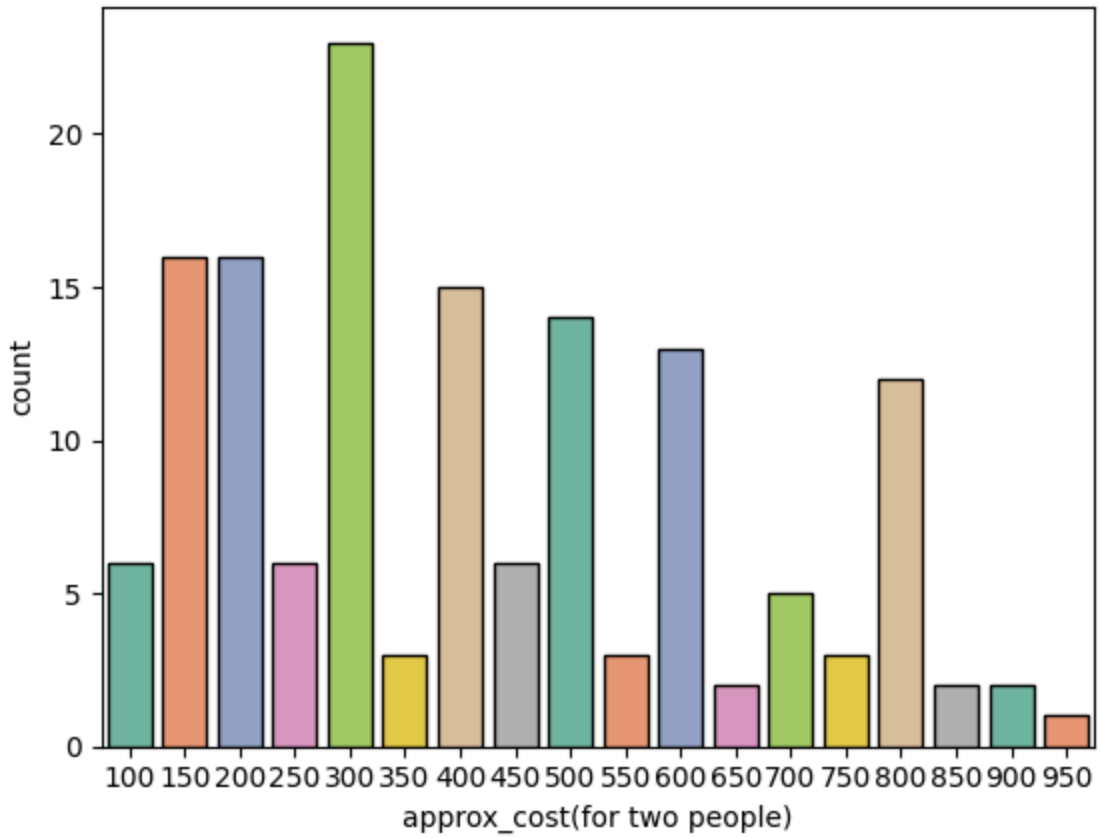
In [28]:
```
# ques  4... zomato has observed that most couples order most of their food online.
```

In [14]:
```
df.head()
```

Out[14]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

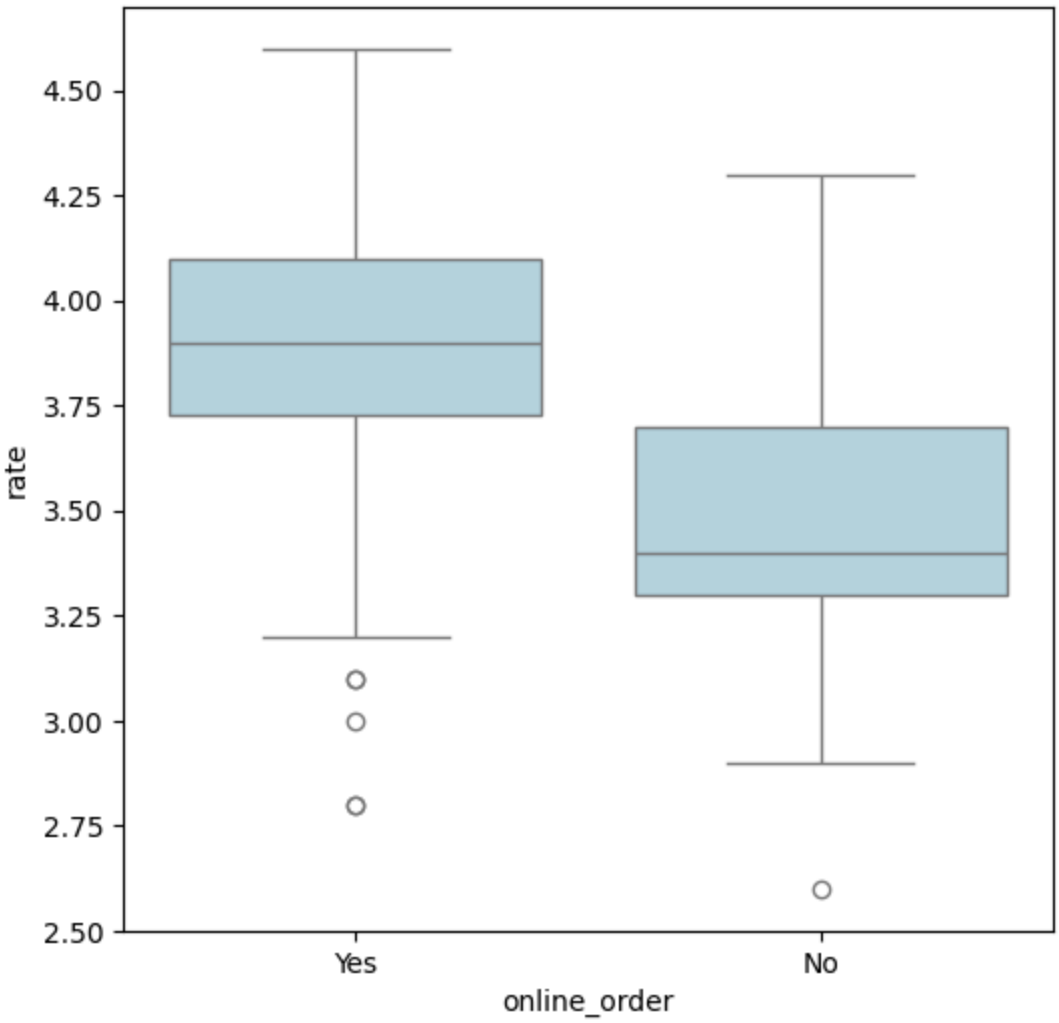
In [50]:
```
couple_data=df['approx_cost(for two people)']
sns.countplot(x=couple_data, data=df,hue='approx_cost(for two people)',palette='Set
plt.legend([],[],frameon=False)
plt.show()
```



In [17]:
```
#  conclusion...  the majority of couples prefer resturants with an approx cost of
```

In [18]:
```
# ques 5...which mode(online or offline) has received the maximum ratings...
```

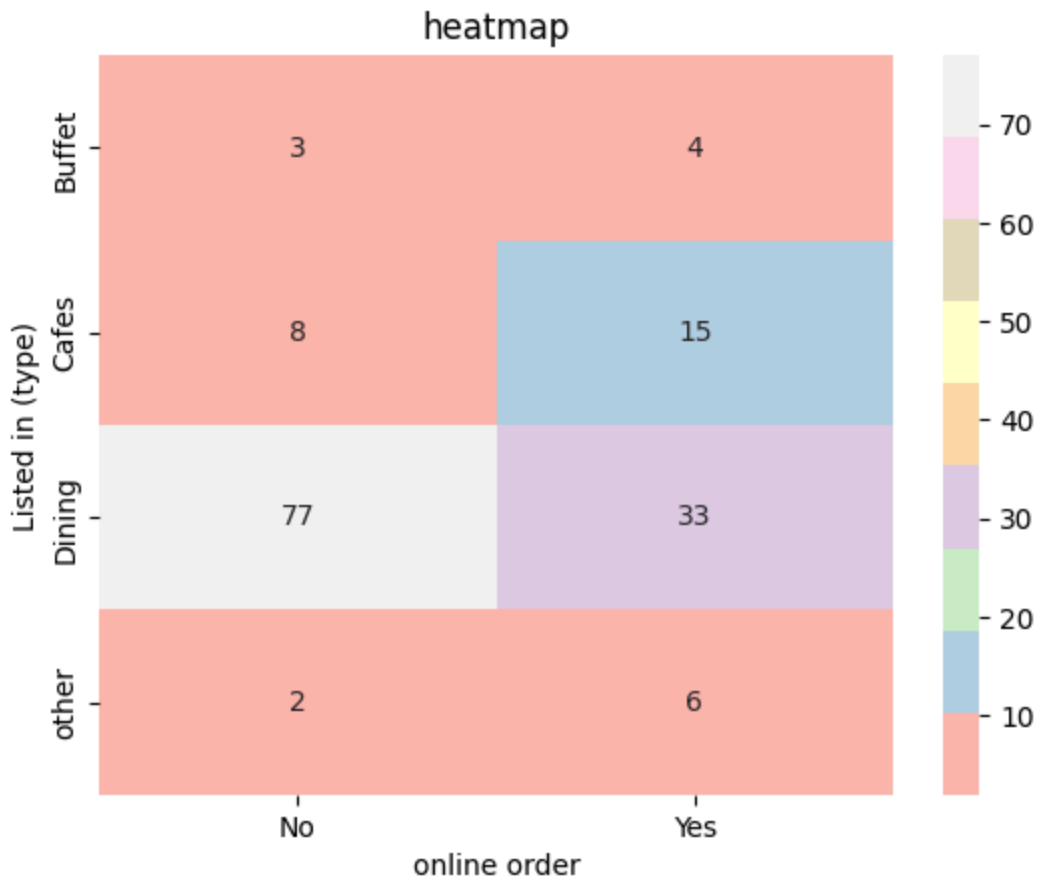
In [69]:
```
plt.figure(figsize=(6,6))
sns.boxplot(x='online_order',y='rate', data=df,color='lightblue')
plt.show()
```



In [21]:
```
#  conclusion... offline order received lower ratings in comparison to online order
```

In [22]:
```
# ques5... which type of resturant received more offline orders,so that zomato can
```

In [32]:
```
pivot_table=df.pivot_table(index='listed_in(type)',columns='online_order',aggfunc='
sns.heatmap(pivot_table,annot=True,cmap='Pastel1',fmt='d')
plt.title('heatmap')
plt.xlabel('online order')
plt.ylabel('Listed in (type)')
plt.show()
```



In [27]:
```
#  conclusion... dining resturants primiarly accept offline orders,whereas cafes pr
# clients preffer orders in person at resturants, but prefer online ordering at caf
```

In [ ]: